Symmetries in Physics — Exercise Sheet 8

Matrix Lie groups and the exponential

Variant: Questions only

Exercise 1 Prove that $\mathrm{GL}(n,\mathbb{C})$, $\mathrm{SU}(n)$, $\mathrm{SL}(n,\mathbb{C})$ and $\mathrm{SL}(n,\mathbb{R})$ are path-connected. *Guidance:* connect a unitary to $\mathbb{1}$ through its spectral decomposition; reach a general invertible matrix by polar decomposition (Exercise 5); for the special linear groups, use that transvections $\mathbb{1} + \lambda e_{ij}$ generate them.

Exercise 2 Compute e^{tX} by hand for (i) the nilpotent $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and (ii) a diagonalisable 2×2 matrix, and verify $\det e^X = e^{\operatorname{tr} X}$ in both cases. Why does the identity then hold for *every* square matrix?

Exercise 3 The symplectic group $\mathrm{Sp}(2,\mathbb{R})$ consists of real 2×2 matrices with $A^{\mathsf{T}} J A = J$, where $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Show that $A^{\mathsf{T}} J A = (\det A)\, J$ for *every* 2×2 matrix, conclude $\mathrm{Sp}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})$, and compute the Lie algebra.

Exercise 4 Fix $\alpha \notin \mathbb{Q}$ and consider $H = \{(e^{\mathrm{i}t}, e^{\mathrm{i}\alpha t}) : t \in \mathbb{R}\} \subset \mathrm{U}(1) \times \mathrm{U}(1)$. Show H is a subgroup but is *not* closed: its closure is the whole torus. (Pigeonhole the fractional parts $\{n\alpha\}$.)

Exercise 5 (i) Show that every positive-definite Hermitian matrix M has a unique positive square root. (ii) Deduce the polar decomposition: every $A \in \mathrm{GL}(n,\mathbb{C})$ factors uniquely as $A = UP$ with U unitary, P positive. (Lecture 14 will spend this on the Lorentz group.)

Exercise 6 A rigid body's orientation is a curve $R(t) \in \mathrm{SO}(3)$. Show that $\Omega(t) = \dot{R} R^{-1}$ is always antisymmetric; identify the angular velocity $\vec{\omega}$ via $\Omega x = \vec{\omega} \times x$; and interpret $R^{-1}\dot{R}$ as the same object seen from the body frame.